Core Web Vitals & Web Performance Checklist

Actionable Engineering Audit for Sub-2s LCP, Low INP, Zero CLS & Edge Caching

1. Largest Contentful Paint (LCP < 2.0 Seconds)

Largest Contentful Paint measures how quickly the main hero headline or visual asset becomes visible in the viewport. Over 70% of poor LCP scores come from unoptimized server rendering, render-blocking fonts, or oversized images.

- [ ] **Pre-Render Above-the-Fold HTML on the Server (SSG / ISR)**
  Ensure the hero headline, availability badge, and primary CTAs are present in the initial HTML document with opacity: 1-never gated behind client-side hydration.
- [ ] **Responsive Image Sizing & Priority Hints**
  Use next/image with accurate sizes="(min-width: 1024px) 33vw, (min-width: 640px) 50vw, 100vw" and AVIF/WebP formats. Set priority on above-the-fold hero imagery only.
- [ ] **Self-Hosted Variable Fonts via next/font**
  Load Inter, Outfit, or JetBrains Mono with display: "swap" and preload: true to eliminate external DNS lookups to fonts.googleapis.com and prevent Flash of Invisible Text (FOIT).

2. Cumulative Layout Shift (CLS < 0.05)

- [ ] **Explicit Aspect-Ratio Containers on Media**
  Wrap all project thumbnails and screenshots in aspect-[16/9] or fixed-dimension containers so the browser reserves exact layout space before images arrive.
- [ ] **Stable Sticky Navigation Chrome**
  Use fixed heights (h-16) on top navigation headers and transform-based animations (translate-y) on mobile bottom bars so scrolling never triggers layout reflow.

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3. Interaction to Next Paint (INP < 150ms)

- [ ] **Keep Client Component Bundles Lean**
  Push static sections (layouts, grids, MDX case studies) to React Server Components and isolate interactivity (modals, carousels, theme toggles) into small leaf Client Components.
- [ ] **Throttle Scroll & Resize Listeners with requestAnimationFrame**
  Always attach scroll listeners with { passive: true } and batch state updates inside window.requestAnimationFrame to maintain 60fps scrolling on mobile devices.
- [ ] **Replace Heavy Tag Managers with Lightweight Analytics**
  Use Plausible (< 1 KB script) or Vercel Analytics instead of multi-hundred-kilobyte third-party tracking bundles that block the main thread.

4. Backend, Database & Edge Delivery (Laravel / Next.js)

- [ ] **Eliminate N+1 Database Queries**
  Use eager loading (Eloquent with() in Laravel or relation includes in Prisma) and index foreign keys + date-range columns used in scheduling and filtering.
- [ ] **Edge Caching & Cloudflare Bot Hygiene**
  Serve static assets with immutable Cache-Control headers. When using Cloudflare in front of Vercel, ensure static chunks (/_next/*) and verified crawlers are never blocked by JS challenges.

Want a full performance & accessibility audit on your existing site? Explore the Conversion Refresh package at https://amilemia.dev/en/services